Bayesian Graphical Lasso for Estimating Brain Networks

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Motivation

We are interested in viewing the brain as a graph i.e. a set of nodes, say regions of the brain, and edges, the connections between these edges

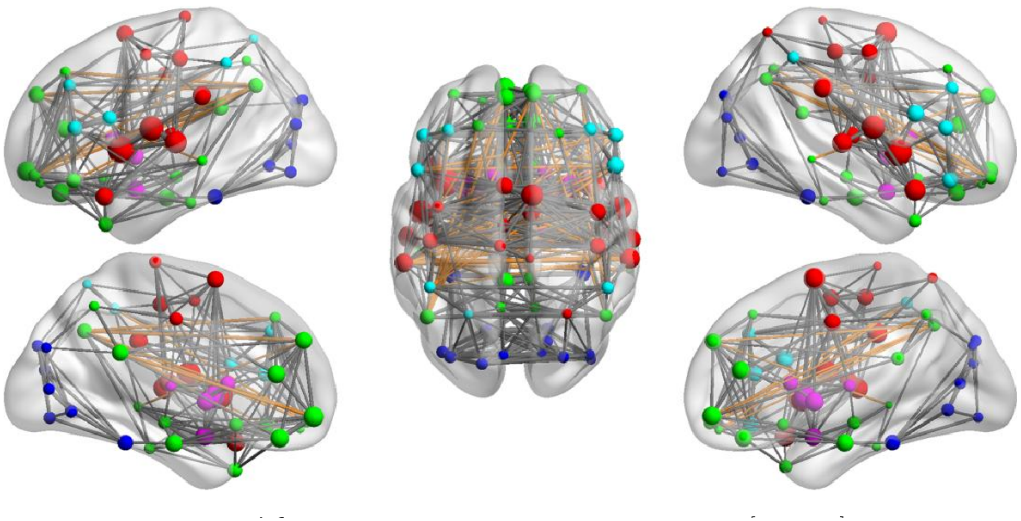


Figure 1: [@Hendricksen2015VisualizingDB]

Different brains have different network structures and therefore different local and global properties of these networks

Motivation

In general, healthy brains have a so-called "small-world" or "scale-free" structure.

Each node has few neighbors but it is easy for one group of nodes or sub-network to talk to another (six degrees of separation).

This implies the existence of hub nodes, nodes through which other areas of the network communicate.

Motivation

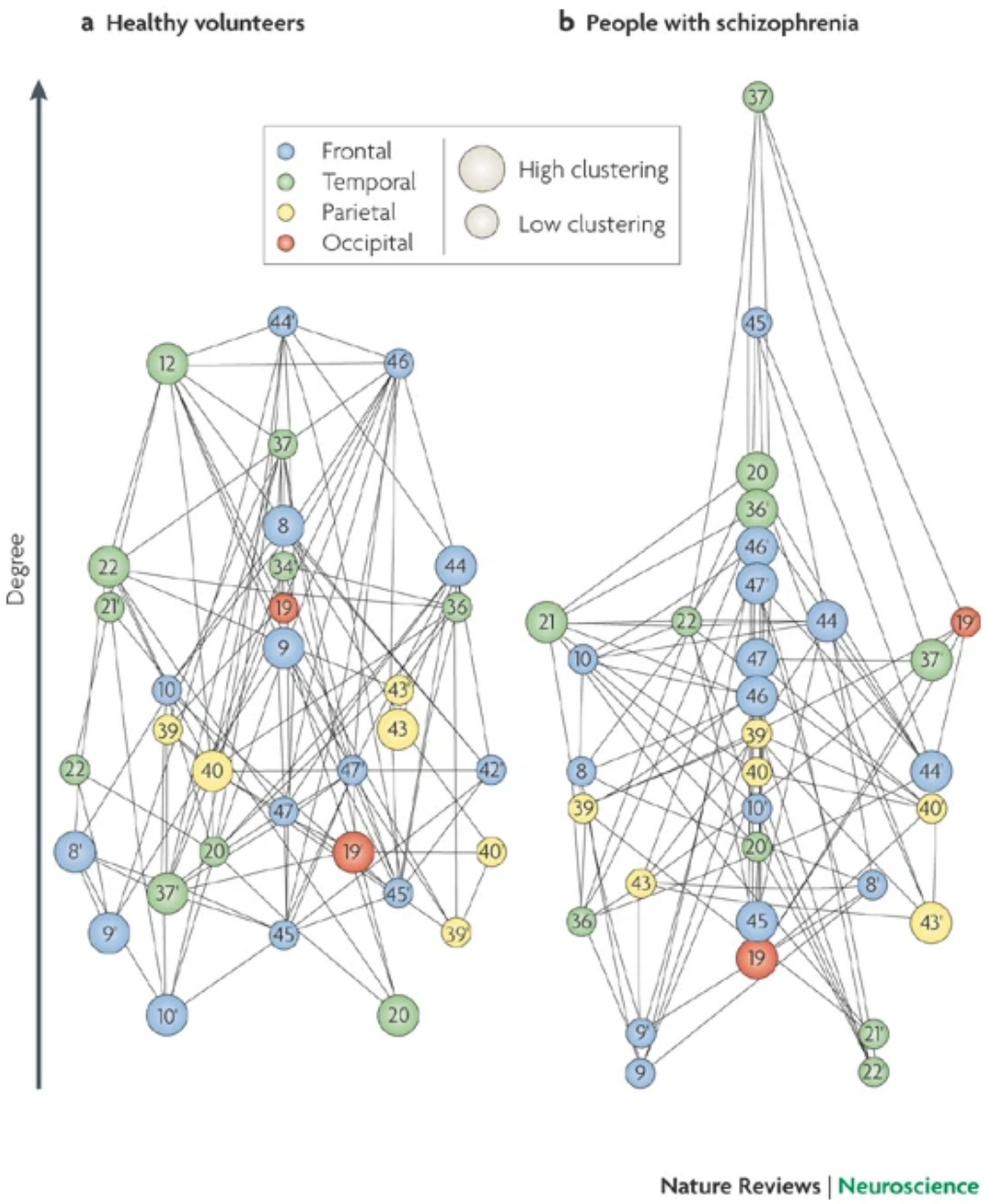
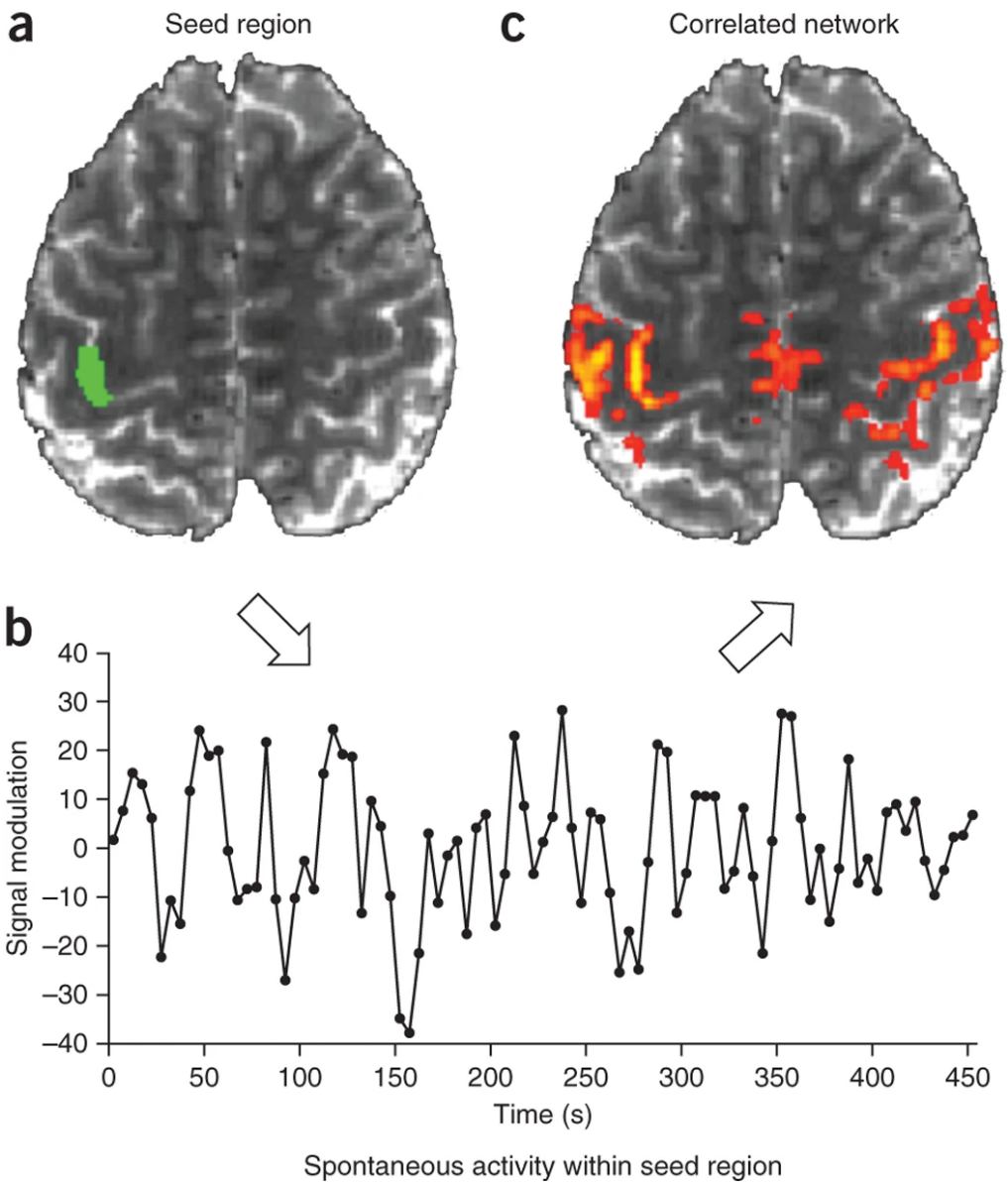


Figure 2: [@Bullmore2009]

Graph Estimation

We don't observe these graphs directly



Graph Estimation

A graph can be represented as an adjacency matrix: a symmetric matrix A with $a_{ij} = 1$ if an edge is present between node i, j and $a_{ij} = 0$ otherwise.

For our purposes, we assume that at time step (i.e. scan) t, the value of the BOLD signal across all D nodes $Y_D \sim MVN(0, \Sigma)$.

Fact: The precision matrix $\Omega = \Sigma^{-1}$ has the property that its entries ω_{ij} encode the partial correlation between node i and node j.

If $\omega_{ij} = 0$, these nodes are conditionally independent.

Graphical Lasso

- To capture this sparsity, something like LASSO (Tibshirani 1996) could work.
- We could fit successive pairwise LASSO's and regress the residuals to obtain partial correlations, but this isn't ideal
- Instead, let's maximize the penalized multivariate gaussian log-likelihood with respect to the precision matrix:

$$\log\,(\det\,\Omega) - \mathrm{tr}(\frac{S}{n}\Omega) - \rho||\Omega||_1$$

- where S is the sample covariance matrix and $||\Omega||_1$ is the entry-wise l_1-norm of Ω.

Graphical Lasso

- This is the approach of Friedman, Hastie, and Tibshirani (2007)
- At a high level, it recursively fits a modified LASSO in a blockwise fashion using sub-gradient methods (gets around non-differentiability of l_1 norm.)
- The focus of my project was a Bayesian approach to the Graphical Lasso by (**10.1214/12-BA729?**) to estimate the aforementioned graphs

Bayesian Graphical Lasso

We have the likelihood and prior

$$p(\mathbf{y}_t|\Omega) = \mathrm{N}(\mathbf{y}_t|0, \Omega^{-1})$$

$$p(\Omega|\lambda) = C^{-1} \prod_{i<j} \mathrm{DE}(\omega_{ij}|\lambda) \prod_{i=1}^{p} \mathrm{Exp}(\omega_{ij}|\frac{\lambda}{2}) \cdot 1_{\Omega \in M^+}$$

where M^+ is the space of positive-definite matrices.

Friedman, Jerome, Trevor Hastie, and Robert Tibshirani. 2007. "Sparse Inverse Covariance Estimation with the Lasso." https://arxiv.org/abs/0708.3517.